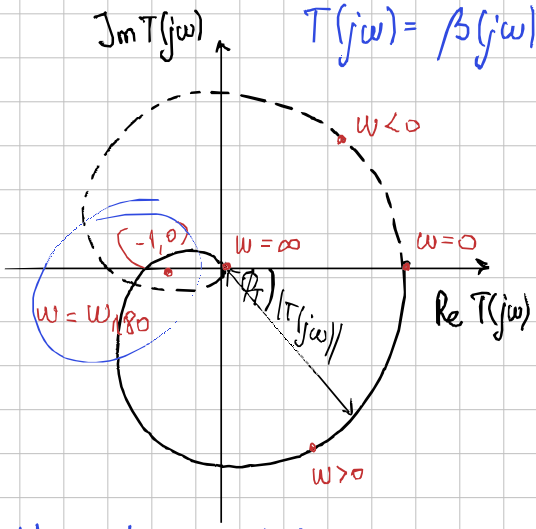
Nyquistov dijagram i stabilnost

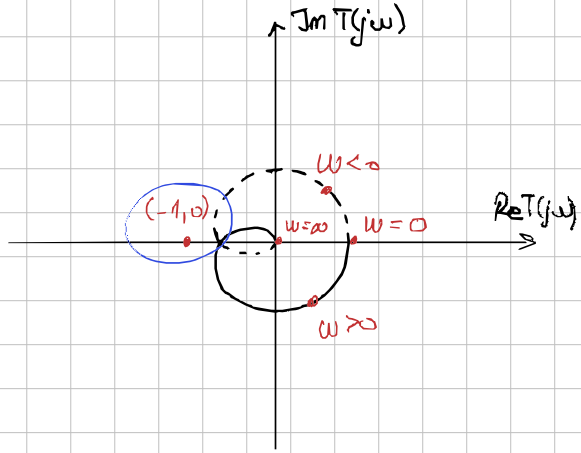
NYQUISTOV DIJAGRAM

Za ispitivanje STABILNOSTI dovoljno je analizirati poj. u petlji P.V.

$T(j\omega) = \beta(j\omega) \cdot A(j\omega)$



Nestabilno pojačalo



Stabilno pojačalo

→ dovoljno je da gledamo točku zakreta $\omega = \omega_{180}$

↳ na frekvenciji $\omega_{180} \rightarrow \phi_T = -180°$

- Za $|T(j\omega_{180})| < 1 \longrightarrow$ STABILNO ne obuhvaća točku (-1,0)
- Za $|T(j\omega_{180})| \geq 1 \longrightarrow$ NESTABILNO → obuhvaća točku (-1,0)

Primjer 7.2)

$$A(s) = \left(\frac{10}{1+\frac{s}{10^4}}\right)^3 \qquad \omega_{180} = ? \qquad T = \beta A, \ \phi_T = -180^\circ$$

$\hookrightarrow \omega_p = 10^4$ rad/s

$\beta \rightarrow$ nestabilno?

$$T(j\omega) = \beta \cdot A = \beta \cdot \frac{10^3}{\left(1+\frac{j\omega}{\omega_p}\right)^3}$$; budući da imamo trostruki pol: $\phi_T(j\omega) = -3\,\mathrm{arctg}\frac{\omega}{\omega_p}$

Na ω_{180}; $\phi_T = -180^\circ \rightarrow -3\,\mathrm{arctg}\left(\frac{\omega_{180}}{\omega_p}\right) = -180^\circ$

$$\mathrm{arctg}\left(\frac{\omega_{180}}{\omega_p}\right) = 60^\circ \longrightarrow \omega_{180} = 17320.51 \text{ rad/s}$$

$$\boxed{\omega_{180} = \sqrt{3}\times 10^4 \text{ rad/s}}$$

Amplituda pojačanja u PPV

$$|T(j\omega)| = |\beta \cdot A(j\omega)| = \frac{\beta \cdot 10^3}{\left(\sqrt{1+\left(\frac{\omega}{\omega_p}\right)^2}\right)^3} = \frac{\beta \cdot 10^3}{\left(\sqrt{1+(\sqrt{3})^2}\right)^3} = \frac{\beta \cdot 10^3}{8} = |T(j\omega)|$$

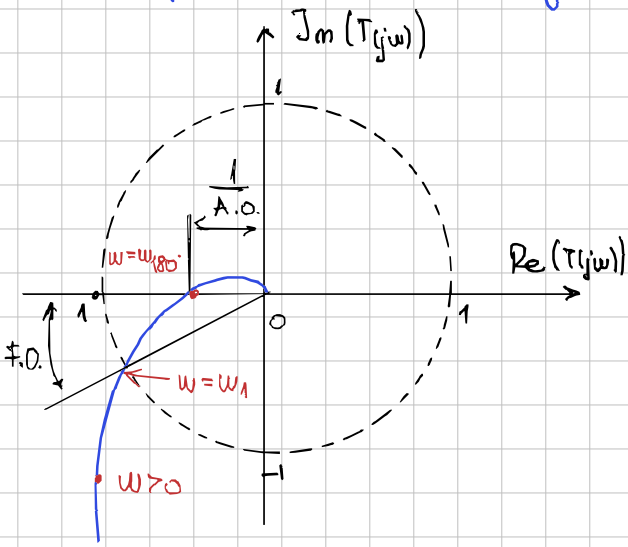
Pojačalo postaje nestabilno uz $|T(j\omega)| = 1$

$$|T(j\omega)| = 1 = \frac{\beta \cdot 10^3}{8} \longrightarrow \beta = \frac{8}{10^3} \longrightarrow \boxed{\beta = 8\times 10^{-3}}$$

$\hookrightarrow$ za svaki $\beta < 0{,}008$ pojačalo je stabilno

—"— $\beta \geq 0{,}008$ —"— nestabilno

Amplitudno i fazno origuranje



Amplitudno origuranje

$$A.O. = \frac{1}{|T(j\omega_{180})|}$$

Fazno origuranje ($\approx 45°$ ugl)

$$F.O. = \phi_T(j\omega_1) + 180°$$

Utjecaj F.O. na frekv. odziv

Pretpostavka: $A_f \approx 1/\beta$

Na frekv jed poj ω_1 $\quad \beta A(j\omega_1) = 1 \cdot \exp(\phi_T)$

$\hookrightarrow \phi_{T_1} = \phi_T(j\omega_1) = F.O. - 180°$

$$A_f(j\omega_1) = \frac{A(j\omega_1)}{1+\beta \cdot A(j\omega_1)} = \frac{1}{\beta} \cdot \frac{\exp(j\phi_{T1})}{1+\exp(j\phi_{T1})}$$

$$|A_f(j\omega_1)| = \frac{1}{|\beta|} \cdot \underbrace{\frac{1}{|1+\exp(j\phi_{T1})|}}_{} \neq A_{0f} \approx \frac{1}{\beta}$$

ne postoj na srednjim frekv → zbog polova i izvoda

Primjer 7.3.) * Poj izvedeno s frekv neovisnim koef. p.V.

? $|A_f(j\omega_1)|$ poj s P.V. ako je poj stabilno s $F.O. = 45°$

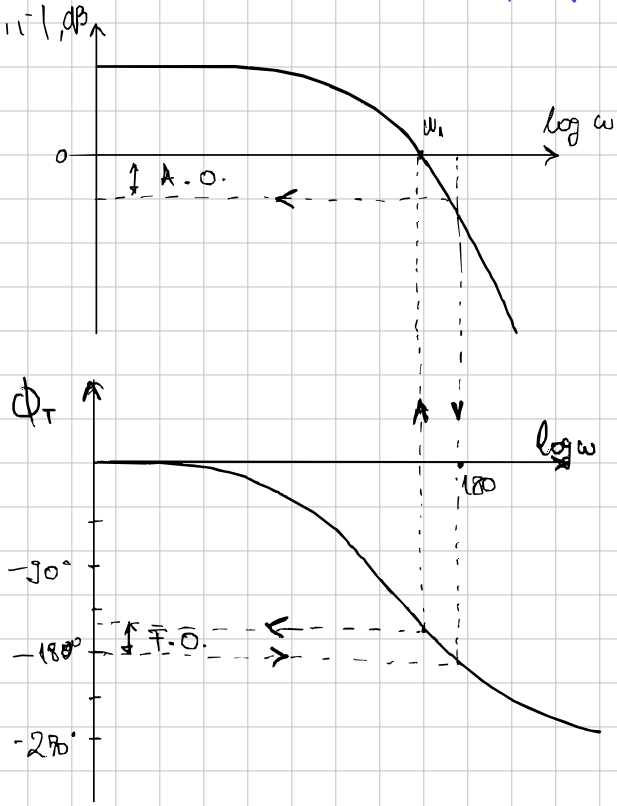
$$|A_f(j\omega_1)| = \frac{1}{|\beta|} \cdot \frac{1}{|1+\exp(j\phi_{T1})|} \longrightarrow \phi_{T1} = F.O. - 180° = 45° - 180$$

$$\phi_{T_1} = -135°$$

$$\exp(j\phi_T) = \cos(\phi_{T1}) + j\sin(\phi_{T1}) = \cos(-135°) + j\sin(-135°)$$

$$|A_f(j\omega_1)| = \frac{1}{|\beta|} \cdot \frac{1}{\sqrt{\left(\frac{2-\sqrt{2}}{2}\right)^2 + \left(\frac{-\sqrt{2}}{2}\right)^2}} = \frac{1}{|\beta|} \cdot 1{,}307$$

Ispitivanje stabilnosti primjenom Bodeovog dijagrama



ovo grafički rješavamo

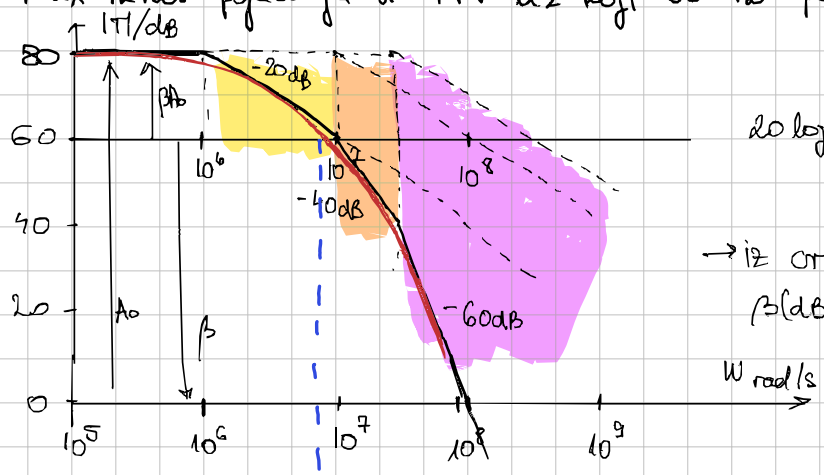
Primjer 7.4.) U PPV primjenomo fju: $A(j\omega) = \dfrac{-10^4}{\left(1+\frac{j\omega}{10^6}\right)\left(1+\frac{j\omega}{10^7}\right)\left(1+\frac{j\omega}{3\times 10^8}\right)}$

β - neovisan o frekv.

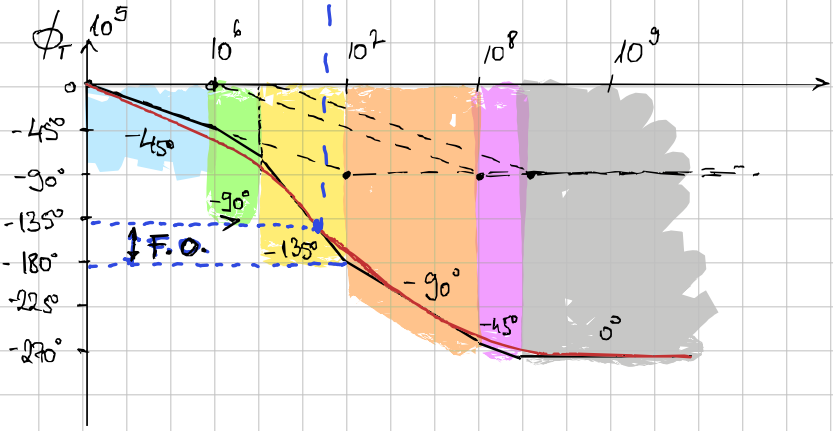
Max iznos pojačanja u PPV uz koji će sustav postići trajno osciliranje

$F.O. = 45°$

$20\log_{10}(10^4) = 80$



→ iz crteža vidimo da je $\beta(dB) < 0$ → spušta se.

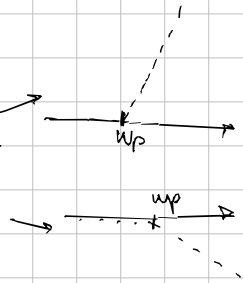


Primjer 7.5.) U PPV prijenosne funkcije osnovnog pojačala i koeficijent P.V. su:

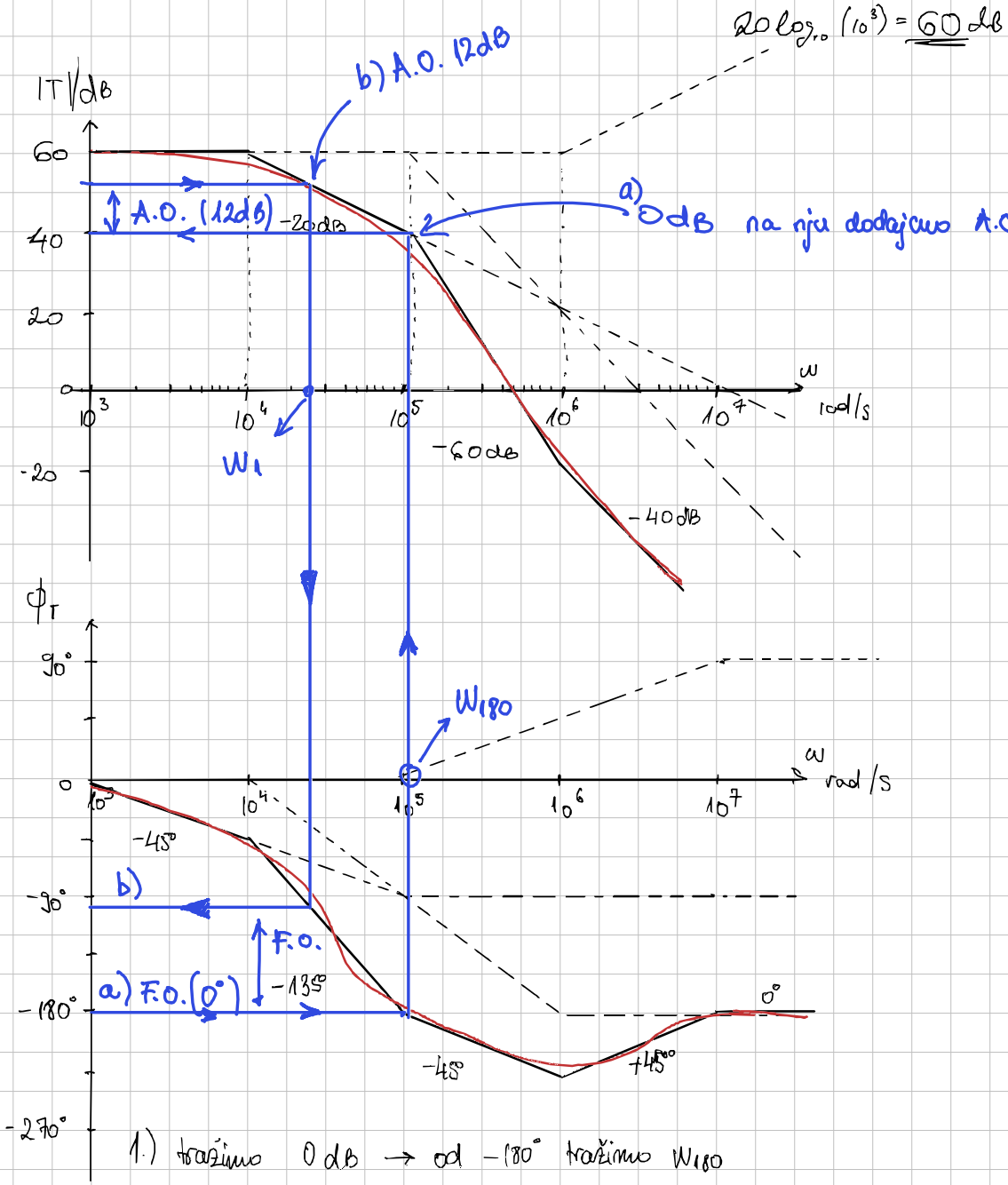
$$A(j\omega) = \frac{-10^3}{\left(1+j\frac{\omega}{8\times10^5}\right)^2} \qquad \beta(j\omega) = \beta_0 \cdot \frac{1+j\frac{\omega}{8\times10^6}}{1+j\frac{\omega}{8\times10^4}}$$

a) $\beta_0 = ?$ Uz F.O. $= 0°$?

$$T = A(j\omega)\cdot\beta(j\omega) = \underbrace{\beta_0 \frac{-10^3}{\left(1+j\frac{\omega}{8\times10^5}\right)^2}}_{\text{konst.}} \cdot \frac{\left(1+\frac{j\omega}{8\times10^6}\right)}{\left(1+j\frac{\omega}{8\times10^4}\right)}$$



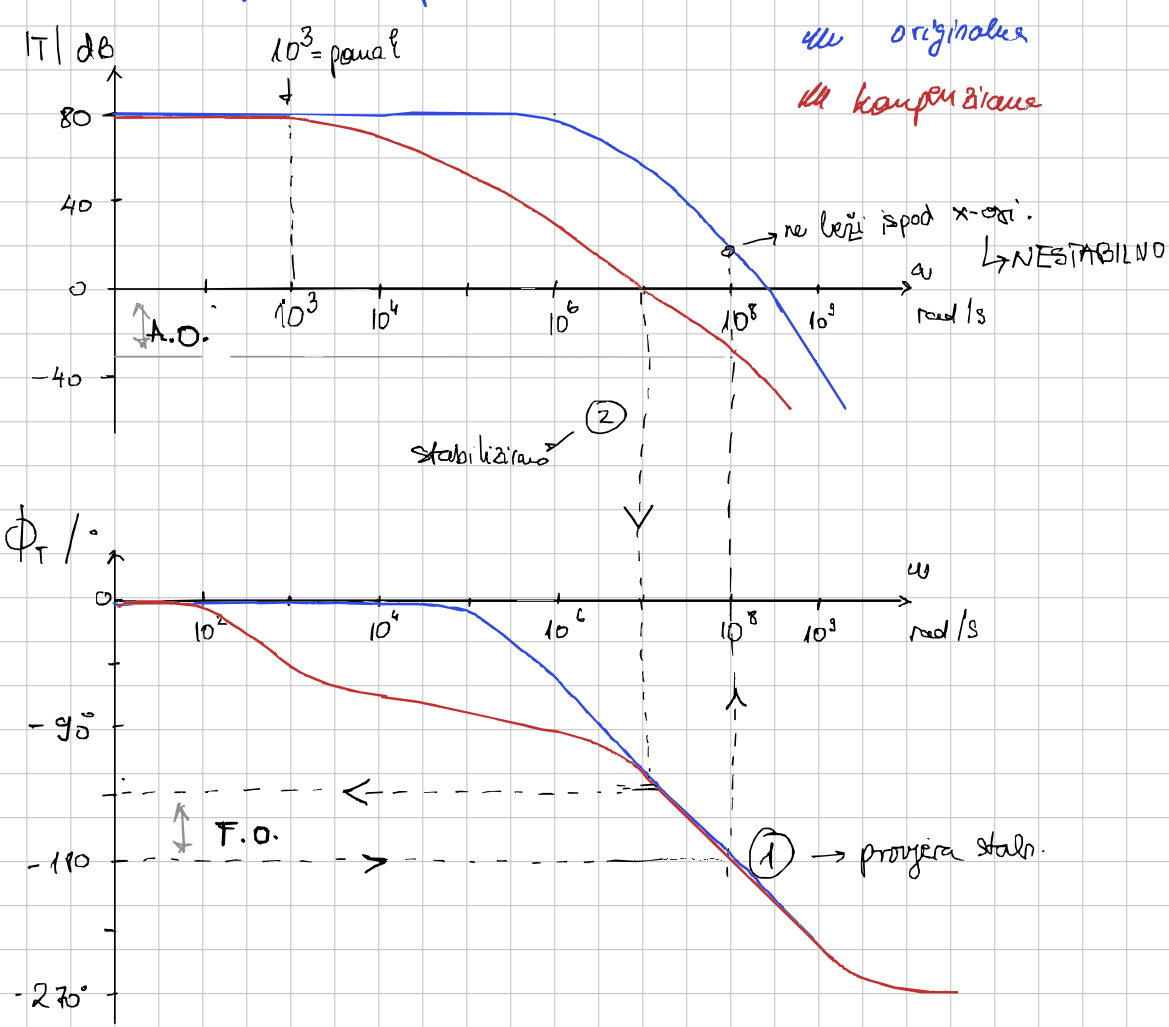
$20\log_{10}(10^3) = \underline{\underline{60\,dB}}$



1.) tražimo 0 dB → od −180° tražimo ω_{180}

2) od 0dB dodajemo A.O (ovdje 12dB)

Frekvencijska kompenzacija dominantnim polom



ORIGINAL. KOMPENZ.

$\omega_1 = 10^6$ rad/s ⟶ 10^3 rad/s (stavljen veći kond, i rješimo pol)

$\omega_2 = 10^8$ rad/s

$\omega_3 = 10^8$ rad/s